

Full Stack Developer Assignment

Overview

Build a mini payment gateway. Merchants should be able to sign up, accept payments, issue refunds, and receive notifications when payment status changes.

Tech Stack

- Backend: NestJS with TypeScript
- Frontend: React with TypeScript
- Database: PostgreSQL
- Version Control: Git (GitHub)

Functional Requirements

Merchants

Merchants should be able to register on the platform, receive the credentials needed to use the gateway, and log in to their dashboard.

Payments

Merchants should be able to create a payment request in the system on behalf of their customer. The customer should then be able to complete the payment through a separate flow, which updates the payment status accordingly. Each payment should carry its amount, customer information, and a status that reflects its lifecycle.

Payments should be retrievable later, and merchants should be able to view a list of all their payments with filtering.

Duplicate payment requests should not create duplicate transactions.

Refunds

Merchants should be able to refund a payment, fully or partially. Refunds must not exceed the original payment amount. The payment's status should reflect its refund state.

Notifications

When a payment's status changes, the merchant should be notified automatically. Failed notifications should be retried. All delivery attempts should be tracked.

Dashboard

A web dashboard for merchants showing:

- Summary of payment activity
- List of payments with filters
- Detailed view of any payment, including its refunds and notification history
- Ability to create a test payment request and simulate its completion
- Ability to issue a refund

The dashboard must work properly on mobile, tablet, and desktop.

Non-Functional Requirements

- Money must be handled accurately.
- Only authenticated merchants should access their own data.
- All inputs must be validated with meaningful error responses.
- Write meaningful tests for the critical paths.
- README should cover setup, key design decisions, trade-offs, and future improvements.

Submission

- Push the code to a public GitHub repository.
- Host both the API and the web dashboard. Share the live URLs along with the repository link.

A follow-up walkthrough will be scheduled after review.